COF and wear of 304 stainless steel in dry reciprocating sliding

ME 323, IIT Bombay - final report (generated by run_all.py)

23 runs, ball-on-flat (Ø 6 mm bearing-steel ball), A = 5 mm, L 10–100 N, f 4–15 Hz, t 60–120 s. Generated 2026-10-27 17:16.

target	final model	selected ML model	R^2 P1	R^2 P2
COF, first 10 s (primary)	$\mu_0 + c \cdot L^{(-1/3)}$	ANN + physics	+0.36	+0.23
COF, end of run (second-half median)	$\mu_0 + c \cdot L^{(-1/3)}$	none	+0.08	+0.05
log10 wear volume V [mm³]	Archard (slope 1)	ANN	+0.29	+0.15
log10 track width [µm]	a + b·ln L	GPR + physics	+0.20	+0.39
log10 k [mm³/(N·m)]	Archard constant k	ANN + physics	-0.15	-0.20

Data summary

run_id	role	replicate_group	test_date	Load_N	Freq_Hz	Run_time_s	cycles	S_m	COF_first10s	COF_end	wear_status	wear_location	k_mm3_Nm	k_upper_mm3_Nm	track_width_um
R01	training	R01	2026-10-13	10	6	120	719	14.4	0.141	0.145	detected	a	3.88e-05	–	442
R02	training	R02	2026-10-13	15	4	120	479	9.59	0.159	0.165	detected	a	1.88e-05	–	311
R04	training	R04	2026-10-13	32	7.5	84	628	12.6	0.135	0.152	below_LOD	–	–	6.64e-06	424
R05	training	R05	2026-10-13	10	15	60	896	17.9	0.159	0.153	detected	c	1.13e-05	–	386
R06	training	R06	2026-10-13	24.5	4.5	116	521	10.4	0.135	0.149	below_LOD	–	–	1.07e-05	349
R07	training	R07	2026-10-13	10.5	12.5	68	847	16.9	0.129	0.139	below_LOD	–	–	1.02e-05	238
R08	training	R08	2026-10-13	54	12	72	861	17.2	0.127	0.146	detected	c	5.22e-06	–	491
R09	training	R09	2026-10-13	20	11	62	679	13.6	0.148	0.16	below_LOD	–	–	1.07e-05	335
R10	training	R10	2026-10-13	40.5	6.5	72	467	9.33	0.131	0.139	below_LOD	–	–	7.86e-06	361
R11	training	R11	2026-10-16	100	4	60	238	4.77	0.13	0.14	detected	c	1.34e-04	–	642
R12	training	R12	2026-10-16	100	7.5	120	897	17.9	0.119	0.118	not_measurable	–	–	–	733
R13	training	R13	2026-10-16	16	5.5	82	449	8.98	0.126	0.128	detected	b	5.84e-05	–	266
R14	training	R04	2026-10-16	32	7.5	84	627	12.5	0.125	0.142	detected	a	6.11e-05	–	370
R17	training	R17	2026-10-16	15	6.5	100	648	12.9	0.129	0.132	detected	c	6.11e-05	–	305
R19	training	R19	2026-10-16	89	7.5	68	507	10.1	0.118	0.121	detected	b	4.46e-05	–	618
R20	training	R04	2026-10-16	32	7.5	84	628	12.6	0.126	0.14	detected	b	4.92e-05	–	318
R21	training	R21	2026-10-16	47.5	7	108	753	15.1	0.12	0.126	detected	m	1.78e-05	–	321
R22	training	R22	2026-10-16	80	4	70	278	5.57	0.129	0.129	detected	b	6.53e-05	–	458
R24	test	R24	2026-10-16	23	5	64	318	6.36	0.131	0.131	detected	c	9.80e-06	–	133
R25	test	R25	2026-10-16	12	9	86	770	15.4	0.136	0.131	detected	m	2.51e-05	–	187
R26	test	R26	2026-10-16	73	8	102	813	16.3	0.117	0.129	detected	a	1.49e-05	–	421
R28	test	R28	2026-10-16	64	6	64	382	7.64	0.121	0.13	detected	m	2.89e-05	–	227
R29	test	R29	2026-10-16	13.5	5	108	538	10.8	0.15	0.151	below_LOD	–	–	2.41e-05	264

Missing DOE runs (no folder): R00, R03, R15, R16, R18, R23, R27. Wear: 16 detected, 6 below LOD, 1 not measurable.

Final model cards

Model card: COF, first 10 s (primary)

Final model: μ_0 + c·L^(–1/3) **Selected ML model:** ANN + physics **Equation:** COF first 10 s = μ_0 + c·L^(–1/3)

Selection rule: ANN + physics is the selected ML model (beats the constant in P1 and P2, tied with the best P1 RMSE, hybrid preferred) but does not beat the physics baseline robustly (lower RMSE and not 1-SE-tied in both P1 and P2).

Noise floor: replicate SD 0.0056 (R04,R14,R20); spread over runs / replicate SD = 2.2; n = 23 runs.

param	estimate	ci_lo	ci_hi
mu0	0.0995	0.0864	0.114
c	0.0978	0.0492	0.142

model	R² P1	RMSE P1	R² P2	RMSE P2	R² P3	RMSE P3	cov95 P1 %	perm p
μ_0 + c·L^(–1/3)	0.36	0.00947	0.225	0.0104	0.702	0.00638	95.7	5.00e-04
ANN + physics	0.43	0.00893	0.0385	0.0116	0.813	0.00506	17.4	0.0099
Constant	-0.0939	0.0124	-0.124	0.0126	-0.0196	0.0118	91.3	–

Model card: COF, end of run (second-half median)

Final model: μ_0 + c·L^(–1/3) **Selected ML model:** none **Equation:** COF end = μ_0 + c·L^(–1/3)

Selection rule: required ML models beat the constant in P1 and P2, but none is tied with the best P1 RMSE.

Noise floor: replicate SD 0.0062 (R04,R14,R20); spread over runs / replicate SD = 2.0; n = 23 runs.

param	estimate	ci_lo	ci_hi
mu0	0.116	0.1	0.134
c	0.069	0.0176	0.118

model	R² P1	RMSE P1	R² P2	RMSE P2	R² P3	RMSE P3	cov95 P1 %	perm p
μ_0 + c·L^(–1/3)	0.0782	0.0115	0.052	0.0122	-0.376	0.00973	95.7	0.0215
Constant	-0.0981	0.0126	-0.125	0.0133	-0.487	0.0101	95.7	–

Model card: log10 wear volume V [mm³]

Final model: Archard (slope 1) **Selected ML model:** ANN **Equation:** log10 V = log10 k + log10(L·S)

Selection rule: ANN is the selected ML model (beats the constant in P1 and P2, tied with the best P1 RMSE) but does not beat the physics baseline robustly (lower RMSE and not 1-SE-tied in both P1 and P2).

Noise floor: replicate SD 0.0668 (R14,R20); spread over runs / replicate SD = 7.1; n = 16 runs.

param	estimate	ci_lo	ci_hi
log10_k	-4.53	-4.73	-4.36

model	R² P1	RMSE P1	R² P2	RMSE P2	R² P3	RMSE P3	cov95 P1 %	perm p
Archard (slope 1)	0.289	0.387	0.151	0.412	0.383	0.341	93.8	0.009
ANN	0.737	0.236	0.162	0.41	-0.837	0.589	100	–
Constant	-0.149	0.492	-0.205	0.491	-0.369	0.509	100	–

Model card: log10 track width [µm]

Final model: a + b·ln L **Selected ML model:** GPR + physics **Equation:** log10 w = a + b·ln L

Selection rule: GPR + physics is the selected ML model (beats the constant in P1 and P2, tied with the best P1 RMSE, hybrid preferred) but does not beat the physics baseline robustly (lower RMSE and not 1-SE-tied in both P1 and P2).

Noise floor: replicate SD 0.0625 (R04,R14,R20); spread over runs / replicate SD = 2.7; n = 23 runs.

param	estimate	ci_lo	ci_hi
a	2.11	1.81	2.37
b	0.126	0.0529	0.204
exponent_beta (w ~ L^beta)	0.291	0.122	0.469
Hertz expectation for b	0.145	–	–

model	R² P1	RMSE P1	R² P2	RMSE P2	R² P3	RMSE P3	cov95 P1 %	perm p
a + b·ln L	0.198	0.15	0.393	0.1	-1.54	0.265	91.3	0.005
GPR + physics	0.262	0.144	0.548	0.0866	-1.54	0.265	87	0.00997
Constant	-0.0925	0.175	-0.12	0.136	-1.96	0.286	95.7	–

Model card: log10 k [mm³/(N·m)]

Final model: Archard constant k **Selected ML model:** ANN + physics **Equation:** log10 k = constant (Archard)

Selection rule: ANN + physics beats the constant and, beyond the 1-SE tie, the physics baseline in both P1 and P2, and is tied with the best P1 RMSE; but without its most influential run (R11) it no longer beats the constant in P1, so the gain is not robust.

k as one material constant: 2.96e-05 mm³/(N·m) ×/÷ 2.36 (95 % bootstrap CI of the geometric mean 1.86e-05 – 4.38e-05).

Noise floor: replicate SD 0.0670 (R14,R20); spread over runs / replicate SD = 5.6; n = 16 runs.

param	estimate	ci_lo	ci_hi
log10_k	-4.53	-4.73	-4.36

model	R² P1	RMSE P1	R² P2	RMSE P2	R² P3	RMSE P3	cov95 P1 %	perm p
Archard constant k	-0.149	0.387	-0.201	0.412	-2.34	0.341	93.8	–
ANN + physics	0.399	0.28	0.386	0.295	-4.47	0.437	87.5	0.0198

Comparison studies C1–C6

C1 – Which COF definition?

Verdict: Use COF_first10s: it follows the adhesion law best among the definitions (P1 R^2 +0.36, P2 +0.23; end-of-run COF +0.08) with signal-to-noise 2.2; the instrument mean reads 13 % low and its ratio to ours falls with frequency (r = -0.70).

label	n	mean	replicate_sd	signal_to_noise	physics_R2_P1	physics_R2_P2	c_adhesion
first 10 s (primary)	23	0.132	0.00557	2.17	0.36	0.225	0.0978
0.5-2 m of sliding	23	0.132	0.00665	1.79	0.306	0.175	0.0912
end: second-half median	23	0.139	0.00619	1.98	0.0782	0.052	0.069
last 10 s	23	0.141	0.00603	2.13	0.0314	-0.01	0.0674
whole run (mean)	23	0.137	0.00664	1.75	0.171	0.128	0.0746
instrument |Fx|/Fz mean	23	0.119	0.00577	1.55	-0.138	-0.195	0.0224

C2 – Which wear measure?

Verdict: Use the most-worn valid location with robust levelling: 16 runs measurable with an LOD, replicate SD 0.07 dex, Archard R^2 P1 +0.29 / P2 +0.15; the old mean-of-locations, plain-levelling measure gives 0.34 dex and R^2 +0.08 / +0.01. Exponents: V ~ L^0.96 [0.41, 1.59] S^-0.07 [-1.32, 1.61] (Archard: 1, 1). The TA depth rule scores a higher R^2 (+0.52) only because it drops the shallow part of every groove: runs at <= 25 N keep a median 11 % of their groove, runs at >= 60 N 91 %, which inflates the load exponent to 1.9; it is not a volume.

label	runs_measurable	replicate_sd_dex	signal_to_noise	best_model	best_R2_P1	best_R2_P2	exp_L	exp_L_ci	exp_S	exp_S_ci
V most-worn location, robust levelling (final)	16	0.0668	7.1	physics	0.289	0.151	0.956	[0.41, 1.59]	-0.067	[-1.32, 1.61]
V mean of locations, robust levelling	22	0.48	2.23	physics	0.122	0.0518	1.88	[0.72, 3.18]	-0.0696	[-1.91, 2.02]
V most-worn location, plain levelling	13	0.102	2.95	powerlaw	-0.0726	-0.414	0.38	[-0.06, 0.98]	-0.572	[-1.21, 0.61]
V mean of locations, plain levelling (old)	21	0.336	3.4	physics	0.0813	0.0131	1.91	[0.43, 4.00]	1.17	[-0.95, 4.60]
V most-worn, 2-D box high-pass (anchor implementation)	15	0.0621	6.73	physics	0.19	-0.0737	0.811	[0.27, 1.49]	0.168	[-1.15, 1.72]
V most-worn, TA depth rule	15	0.00884	93.7	powerlaw	0.515	0.319	1.88	[1.20, 2.71]	-0.228	[-1.51, 1.60]
max wear depth	21	0.504	1.04	powerlaw	0.0136	-0.176	–	–	–	–
track width	23	0.0625	2.75	physics	0.198	0.393	–	–	–	–

C3 – Which model?

Verdict: COF first 10 s: final = physics law (selected ML: ANN + physics); COF end: final = physics law (selected ML: none); log10 V: final = physics law (selected ML: ANN); log10 width: final = physics law (selected ML: GPR + physics); log10 k: final = physics law (selected ML: ANN + physics).

target	label	P1_R2	P1_RMSE	P2_R2	P3_R2	n_params	uncertainty	physically_consistent	perm_p	verdict
COF_first10s	Constant	-0.0939	0.0124	-0.124	-0.0196	1	yes	no	–	reference
COF_first10s	Physics	0.36	0.00947	0.225	0.702	2	yes	yes	5.00e-04	physics baseline (final model)
COF_first10s	Power law (OLS)	0.185	0.0107	-0.0137	0.619	4	yes	yes	–	worse than physics
COF_first10s	Ridge	0.15	0.0109	-0.107	0.626	4	no	yes	–	worse than physics
COF_first10s	SVR	-0.207	0.013	-0.116	0.69	23 SV	no	yes	–	no better than constant
COF_first10s	XGBoost	0.0924	0.0113	-0.208	0.446	200 trees, depth 1	no	yes	–	unstable (beats constant in P1 only)
COF_first10s	ANN	0.143	0.011	0.154	0.603	21	yes	yes	–	worse than physics
COF_first10s	GPR	-0.169	0.0128	-0.422	0.0734	5	yes	yes	–	no better than constant
COF_first10s	TabPFN	–	–	–	–	–	–	–	–	not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download)
COF_first10s	ANN + physics	0.43	0.00893	0.0385	0.813	23	yes	yes	0.0099	chosen (selected ML model); not better than physics
COF_first10s	GPR + physics	0.148	0.0109	0.0227	0.645	7	yes	yes	0.0399	worse than physics
COF_first10s	TabPFN + physics	–	–	–	–	–	–	–	–	not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download)
COF_end	Constant	-0.0981	0.0126	-0.125	-0.487	1	yes	yes	–	reference
COF_end	Physics	0.0782	0.0115	0.052	-0.376	2	yes	yes	0.0215	physics baseline (final model)
COF_end	Power law (OLS)	-0.112	0.0127	-0.146	-0.842	4	yes	yes	–	no better than constant
COF_end	Ridge	-0.22	0.0133	-0.251	-0.842	4	no	yes	–	no better than constant

target	label	P1_R2	P1_RMSE	P2_R2	P3_R2	n_params	uncertainty	physically_consistent	perm_p	verdict
COF_end	SVR	0.608	0.00752	0.0358	0.493	21 SV	no	no	–	tied with best (not a required model)
COF_end	XGBoost	-0.128	0.0128	-0.17	-0.969	200 trees, depth 1	no	yes	–	no better than constant
COF_end	ANN	0.396	0.00933	0.00362	0.512	21	yes	no	–	beats constant, not tied with best
COF_end	GPR	0.374	0.0095	-0.383	-0.141	5	yes	no	–	unstable (beats constant in P1 only)
COF_end	TabPFN	–	–	–	–	–	–	–	–	not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download)
COF_end	ANN + physics	0.156	0.011	-0.587	0.62	18	yes	no	0.158	unstable (beats constant in P1 only)
COF_end	GPR + physics	0.203	0.0107	-0.224	-0.531	7	yes	yes	0.0133	unstable (beats constant in P1 only)
COF_end	TabPFN + physics	–	–	–	–	–	–	–	–	not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download)
log10_V	Constant	-0.149	0.492	-0.205	-0.369	1	yes	no	–	reference
log10_V	Physics	0.289	0.387	0.151	0.383	1	yes	yes	0.009	physics baseline (final model)
log10_V	Power law (OLS)	0.193	0.412	0.169	-0.338	4	yes	no	–	worse than physics
log10_V	Ridge	0.159	0.421	0.178	-0.358	4	no	no	–	worse than physics
log10_V	SVR	0.0657	0.444	0.529	-0.571	16 SV	no	no	–	worse than physics
log10_V	XGBoost	0.21	0.408	0.332	0.0351	200 trees, depth 2	no	no	–	worse than physics
log10_V	ANN	0.737	0.236	0.162	-0.837	21	yes	no	–	chosen (selected ML model); not better than physics
log10_V	GPR	0.0196	0.454	0.795	-1.51	5	yes	no	–	worse than physics

target	label	P1_R2	P1_RMSE	P2_R2	P3_R2	n_params	uncertainty	physically_consistent	perm_p	verdict
log10_V	TabPFN	–	–	–	–	–	–	–	–	not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download)
log10_V	ANN + physics	0.629	0.28	0.549	-0.031	22	yes	no	0.0099	beats constant, not tied with best
log10_V	GPR + physics	0.0816	0.44	0.216	-0.413	6	yes	no	0.0498	worse than physics
log10_V	TabPFN + physics	–	–	–	–	–	–	–	–	not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download)
log10_width	Constant	-0.0925	0.175	-0.12	-1.96	1	yes	no	–	reference
log10_width	Physics	0.198	0.15	0.393	-1.54	2	yes	yes	0.005	physics baseline (final model)
log10_width	Power law (OLS)	0.0758	0.161	0.134	-1.31	4	yes	no	–	tied with best (not a required model)
log10_width	Ridge	0.0078	0.167	0.0241	-1.31	4	no	yes	–	worse than physics
log10_width	SVR	0.204	0.15	0.0817	-1.3	22 SV	no	yes	–	tied with best (not a required model)
log10_width	XGBoost	0.102	0.159	0.478	-1.2	400 trees, depth 2	no	no	–	tied with best (not a required model)
log10_width	ANN	-0.191	0.183	-0.116	-1.38	16	yes	no	–	no better than constant
log10_width	GPR	0.166	0.153	0.0017	-1.39	5	yes	no	–	tied with best
log10_width	TabPFN	–	–	–	–	–	–	–	–	not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download)
log10_width	ANN + physics	-0.0169	0.169	0.066	-1.06	18	yes	no	0.0495	worse than physics
log10_width	GPR + physics	0.262	0.144	0.548	-1.54	7	yes	no	0.00997	chosen (selected ML model); not better than physics

target	label	P1_R2	P1_RMSE	P2_R2	P3_R2	n_params	uncertainty	physically_consistent	perm_p	verdict
log10_width	TabPFN + physics	–	–	–	–	–	–	–	–	not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download)
log10_k	Constant	-0.149	0.387	-0.201	-2.34	1	yes	yes	–	reference
log10_k	Physics	-0.149	0.387	-0.201	-2.34	1	yes	yes	–	physics baseline (final model)
log10_k	Power law (OLS)	-0.306	0.412	-0.178	-6.25	4	yes	no	–	no better than constant
log10_k	Ridge	-0.384	0.425	-0.498	-5.17	4	no	yes	–	no better than constant
log10_k	SVR	-0.234	0.401	0.202	-7.2	16 SV	no	no	–	no better than constant
log10_k	XGBoost	-0.493	0.441	0.0305	-5.58	200 trees, depth 2	no	no	–	no better than constant
log10_k	ANN	0.408	0.278	0.365	-4.19	21	yes	no	–	tied with best
log10_k	GPR	-0.485	0.44	-0.109	-6.66	5	yes	no	–	no better than constant
log10_k	TabPFN	–	–	–	–	–	–	–	–	not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download)
log10_k	ANN + physics	0.399	0.28	0.386	-4.47	22	yes	no	0.0198	chosen (selected ML model)
log10_k	GPR + physics	-0.504	0.443	-0.109	-6.66	6	yes	no	0.621	no better than constant
log10_k	TabPFN + physics	–	–	–	–	–	–	–	–	not run (tabpfn installed but cannot load weights: TabPFNLicenseError: TabPFN requires a one-time license acceptance to download)

C4 – Old pipeline vs final?

Verdict: Replace the old pipeline. COF: the old ANN on end-of-run COF scores R^2 +0.40 in P1 but +0.00 on the training runs (P2); the final model on COF_first10s scores +0.36 / +0.23 with two parameters. Wear: R^2 -0.48 / -0.47 (ANN on mean-of-locations k, plain levelling, replicate SD 0.34 dex) -> +0.29 / +0.15 (final model on the most-worn-location volume, replicate SD 0.07 dex).

pipeline	quantity	target	model	n	R2	RMSE	tercile_kappa	replicate_sd	signal_to_noise
old	COF	COF_end (old target)	ANN (plain)	23	0.396	0.00933	0.358	0.00619	1.98
old	wear	log10 k, mean of locations, plain levelling	ANN (plain)	21	-0.484	1.24	-0.214	0.336	3.09
final	COF	COF, first 10 s (primary)	physics	23	0.36	0.00947	0.214	0.00557	2.17
final	wear	log10 wear volume V [mm³]	physics	16	0.289	0.387	0.0698	0.0668	7.1

C5 – Does the literature data help?

Verdict: Do not use the literature data: paper COF ~ L^+0.52 v^-0.69 vs ours L^-0.09 v^-0.05; log10_COF: best with paper data GPR + study flag R^2 +0.24 vs ours alone +0.28; log10_k: best with paper data pooled + study offset: power law R^2 -0.31 vs ours alone +0.41. The published COF (0.002-0.007) and volumes (up to 1620 mm^3 in 200 s) are implausible as printed.

target	strategy	uses_paper	n	R2_P1	RMSE_P1	bias
log10_COF	ours: power law	False	23	0.231	0.033	-0.00149
log10_COF	ours: GPR	False	23	0.0132	0.0374	-0.00577
log10_COF	ours: ANN	False	23	0.278	0.032	-0.00273
log10_COF	naive pooling: power law	True	23	-20.8	0.176	-0.0268
log10_COF	naive pooling: GPR	True	23	-1.55	0.0601	-0.00693
log10_COF	naive pooling: ANN	True	23	-0.212	0.0414	-0.00367
log10_COF	pooled + study offset: power law	True	23	-3.28	0.0778	-0.00146
log10_COF	ours shrunk toward paper slopes	True	23	0.231	0.033	-0.00149
log10_COF	paper slopes imposed	True	23	-53.3	0.277	-7.35e-05
log10_COF	GPR + study flag	True	23	0.237	0.0329	-0.00179
log10_COF	ANN + study flag	True	23	-0.189	0.041	-0.00242
log10_k	ours: power law	False	16	-0.306	0.412	0.0215
log10_k	ours: GPR	False	16	-0.485	0.44	-0.00172
log10_k	ours: ANN	False	16	0.408	0.278	0.00455
log10_k	naive pooling: power law	True	16	-4.1	0.815	0.149
log10_k	naive pooling: GPR	True	16	-0.487	0.44	0.0154
log10_k	naive pooling: ANN	True	16	-0.963	0.506	-0.0178
log10_k	pooled + study offset: power law	True	16	-0.315	0.414	0.0167
log10_k	ours shrunk toward paper slopes	True	16	-0.488	0.44	0.0482
log10_k	paper slopes imposed	True	16	-2.93	0.716	0.00617
log10_k	GPR + study flag	True	16	-0.641	0.462	-0.00524
log10_k	ANN + study flag	True	16	-0.384	0.425	-0.0229

study	quantity	variable	exponent	ci_lo	ci_hi
ours	COF	load	-0.0859	-0.123	-0.0435
ours	COF	speed	-0.0476	-0.151	0.0462
paper	COF	load	0.525	0.0827	0.933

study	quantity	variable	exponent	ci_lo	ci_hi
paper	COF	speed	-0.693	-0.936	-0.444
ours	k	load	0.0218	-0.562	0.739
ours	k	speed	-1.26	-2.62	0.443
paper	k	load	-1.45	-2.1	-0.848
paper	k	speed	-0.334	-0.678	0.0519

C6 – Can the TA script be used?

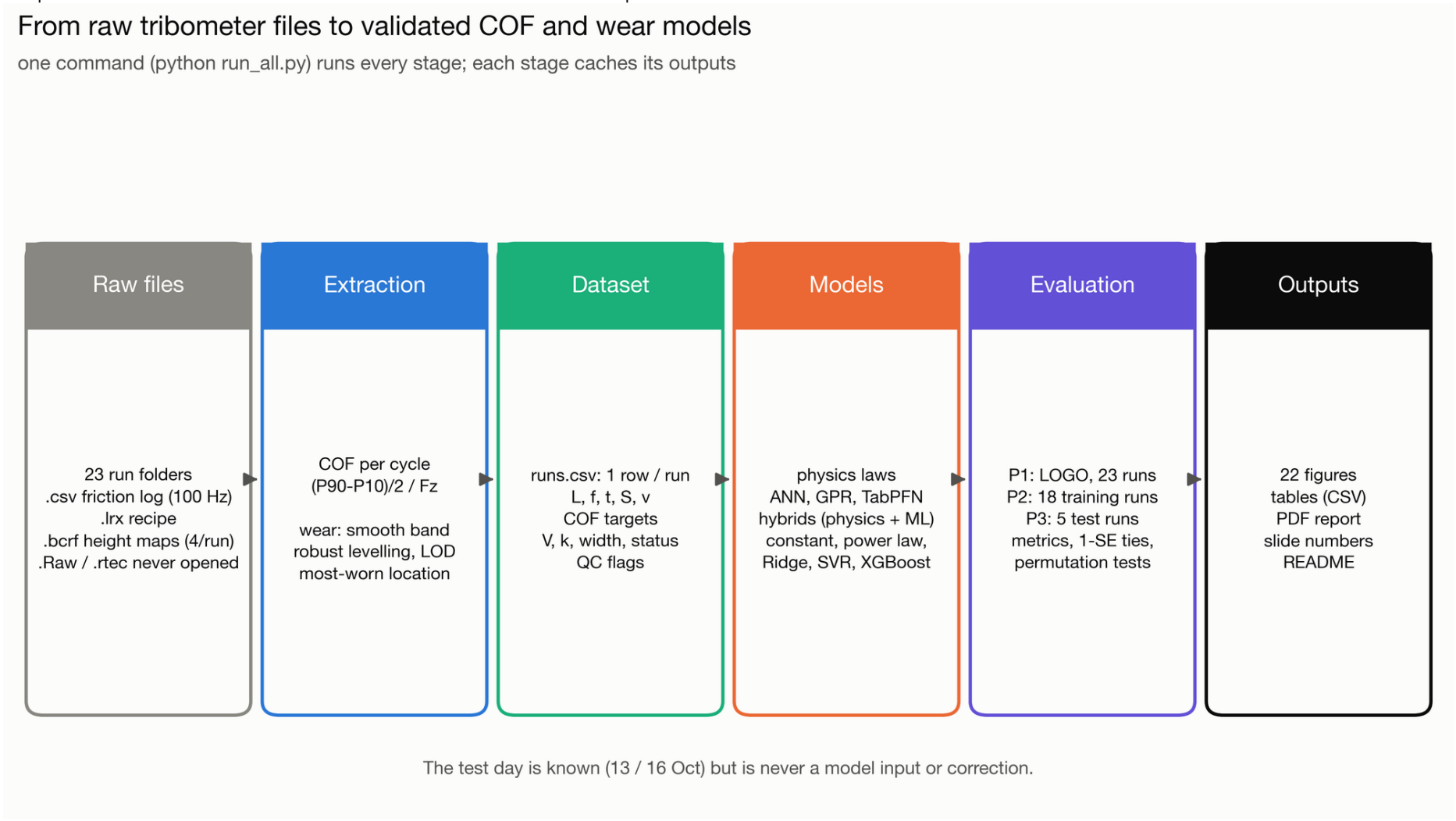
Verdict: Do not use the TA script as written: 25 of 92 scans are read along the track, and the R04/R14/R20 replicates spread 30x in k (SD 0.84 dex); fixed copy 0.13 dex, ours 0.07 dex (R14/R20 only). Archard R^2 (P1): original -0.03, fixed -0.19, ours +0.29.

pipeline	k_R04	k_R14	k_R20	replicate_sd_log10k	replicate_spread_factor	runs_with_volume	archard_R2_P1	cof_law_R2_P1
original TA script	2.33e-06	7.02e-05	2.57e-06	0.841	30.1	21	-0.028	-0.138
fixed TA copy	–	5.62e-05	8.46e-05	0.126	1.51	20	-0.192	0.246
ours	–	6.11e-05	4.92e-05	0.067	1.24	16	0.289	0.36

bug	description	evidence
1	Orientation chosen from the unlevelled map (tilt decides)	25 of 92 scans transposed, so the 'profile' runs along the track. R24.bcrf: transposed, reported depth 22.5 um (a pit), groove area 4352 um^2; with the correct axis the same code gives depth 7.4 um, groove area 0 um^2
2	Points shallower than the cutoff zeroed one by one	10 scans (correct axis) have groove area 0 although the depth is >= 3 um, e.g. R05b_13.bcrf (3.4 um), R11.bcrf (4.2 um), R13.bcrf (4.1 um), R14_a.bcrf (5.9 um)
3	Pile-up summed over the whole image outside the groove	median TA pile-up / band pile-up = 1.5 over 65 valid scans (texture peaks counted as pile-up)
4	Shoulder (reference) rows never checked to be unworn	in 23 of 90 scans with a track, the worn band reaches into the outer 15 % rows that the TA uses as the unworn baseline
5	Scan picked by the deepest point (texture valleys)	the deepest-point scan differs from the largest-groove valid scan in 12 of 20 runs with a detected groove
6	COF = mean of the instrument |Fx|/Fz channel; deceleration second not trimmed	TA COF / offset-free COF = 0.873 on average (13 % low); the ratio falls with frequency (r = -0.70, p = 0.000)
7	Hard-coded R01 parameters; friction log = first *.csv; outputs written into the data folder	R04 with the default arguments (10 N, 6 Hz, 120 s): k = 6.53e-06 instead of 2.33e-06 mm^3/(N m) (2.8x); run_pipeline() writes all_scans_breakdown.csv and single_scan_summary.csv into each run folder (avoided here)

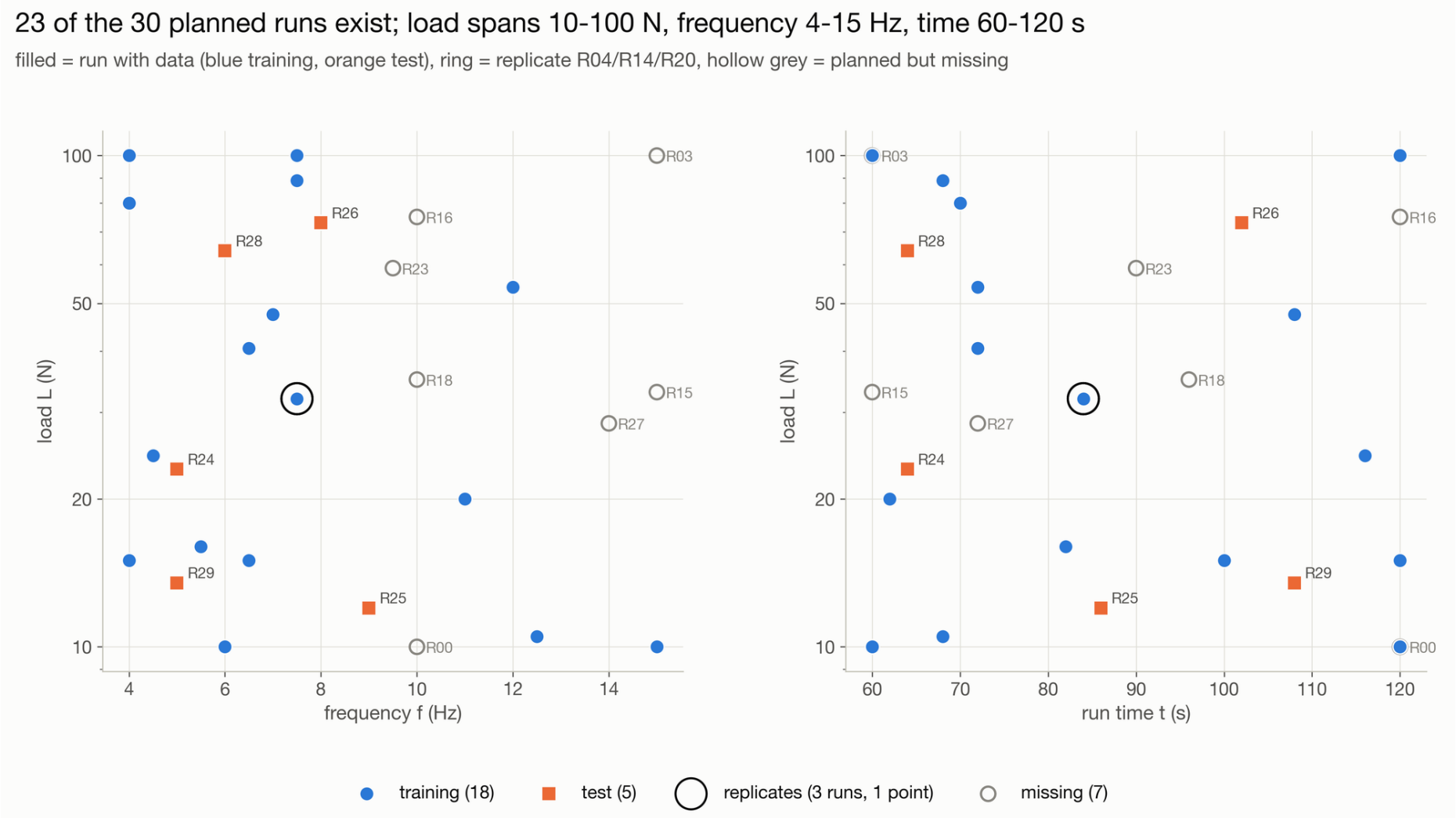
Figures

Figure 1: Pipeline overview: raw files -> extraction -> dataset -> models -> evaluation -> outputs.



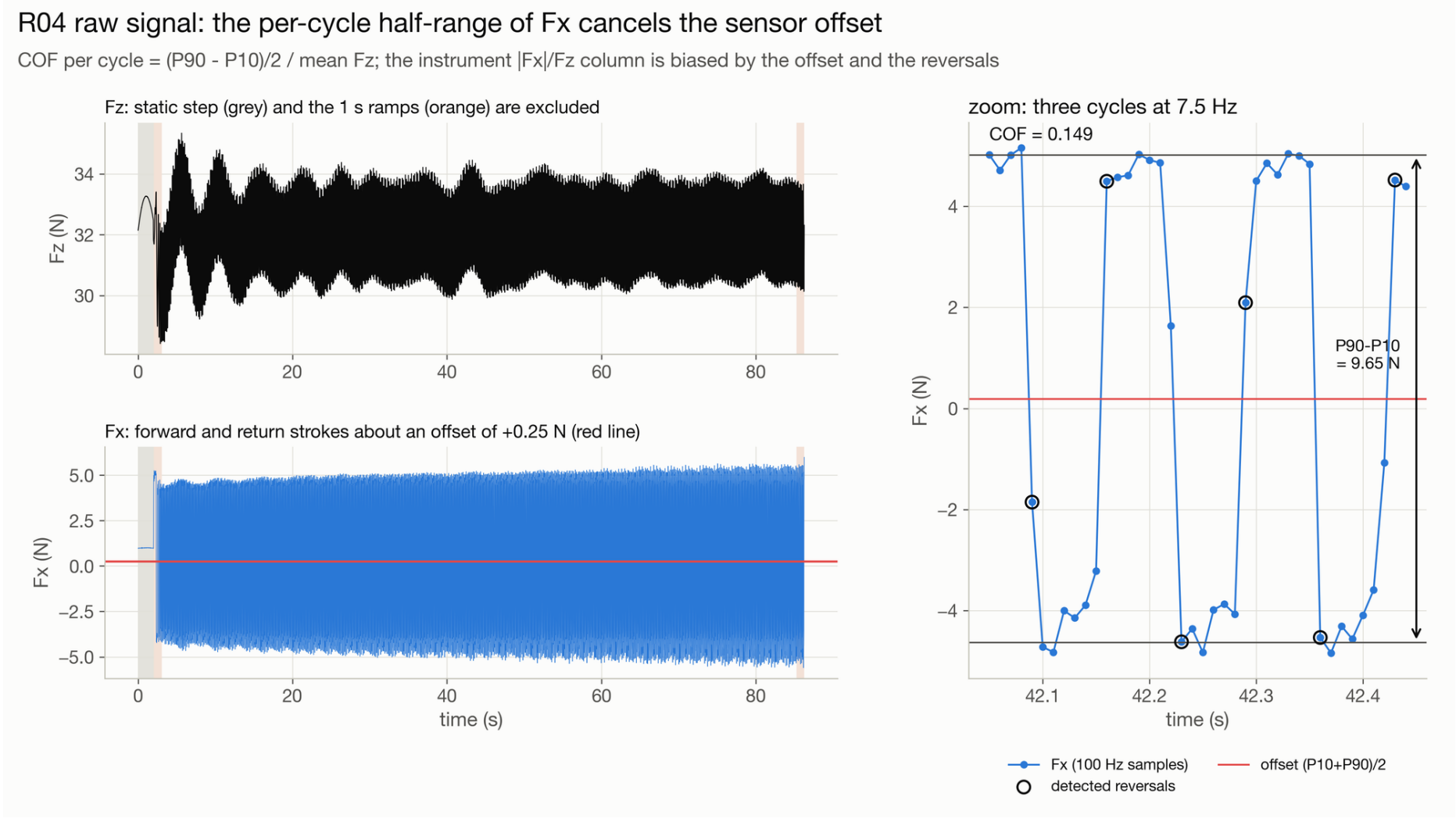
Data: code structure (src/tribo_final)

Figure 2: Test matrix of the 23 runs in L-f-t space; replicates ringed, missing DOE rows hollow.



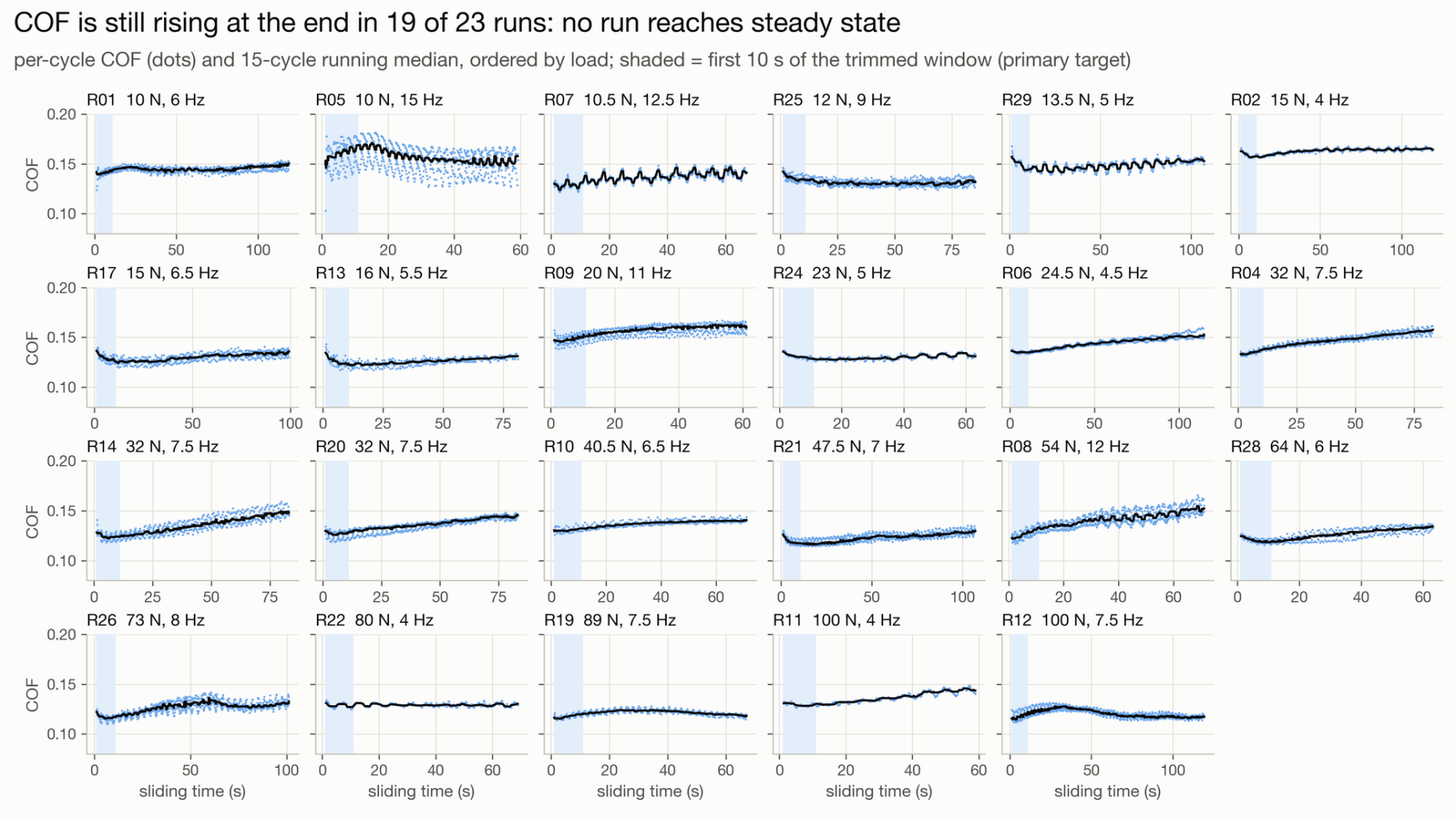
Data: outputs/data/runs.csv, inputs/run_sheet.csv

Figure 3: R04 raw Fz and Fx: static step, trimmed ramps, Fx offset, reversals and the per-cycle half-range COF.



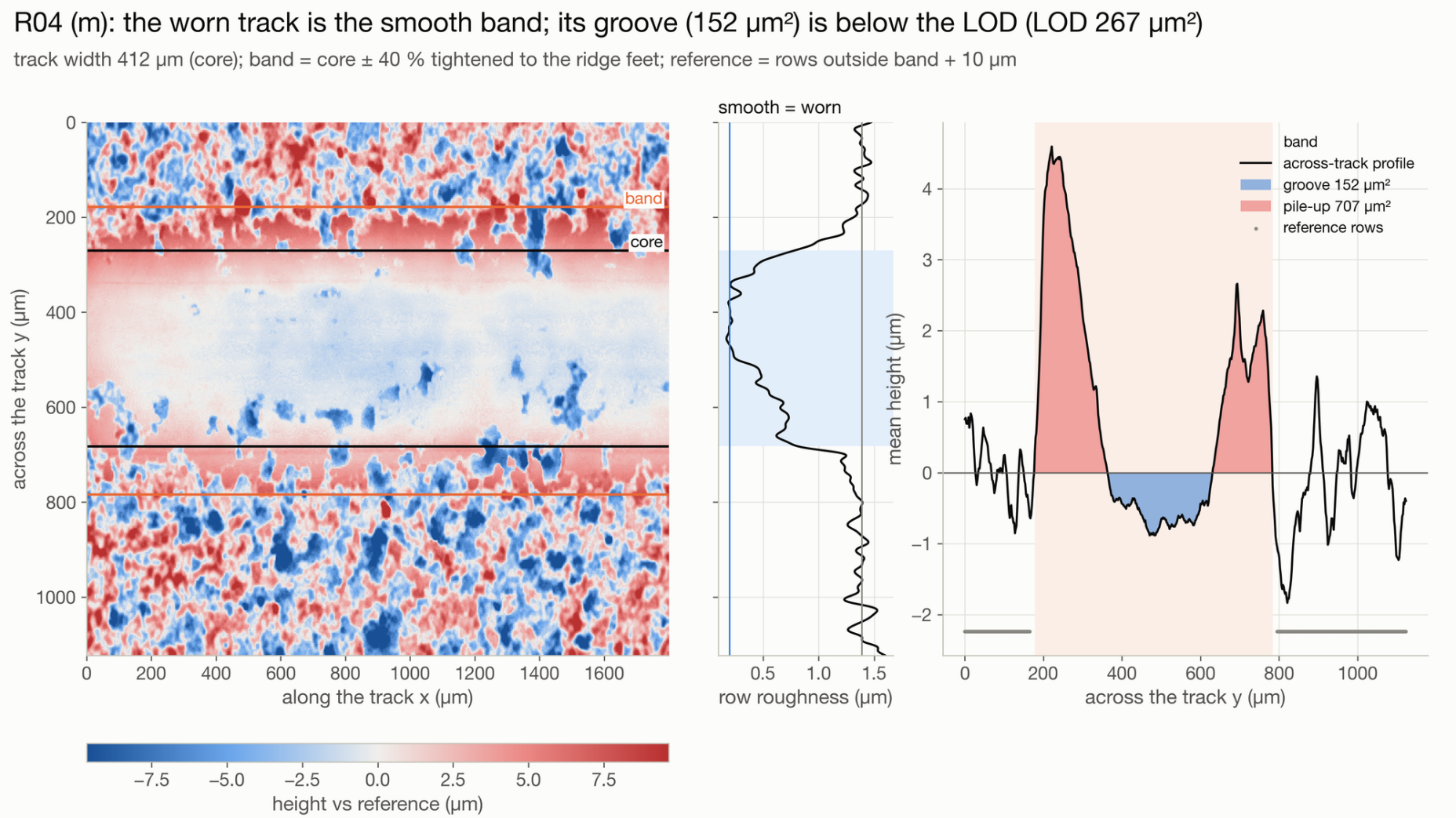
Data: testdata/R04/R04.csv

Figure 4: COF vs sliding time for all 23 runs (small multiples by load); first-10-s window shaded.



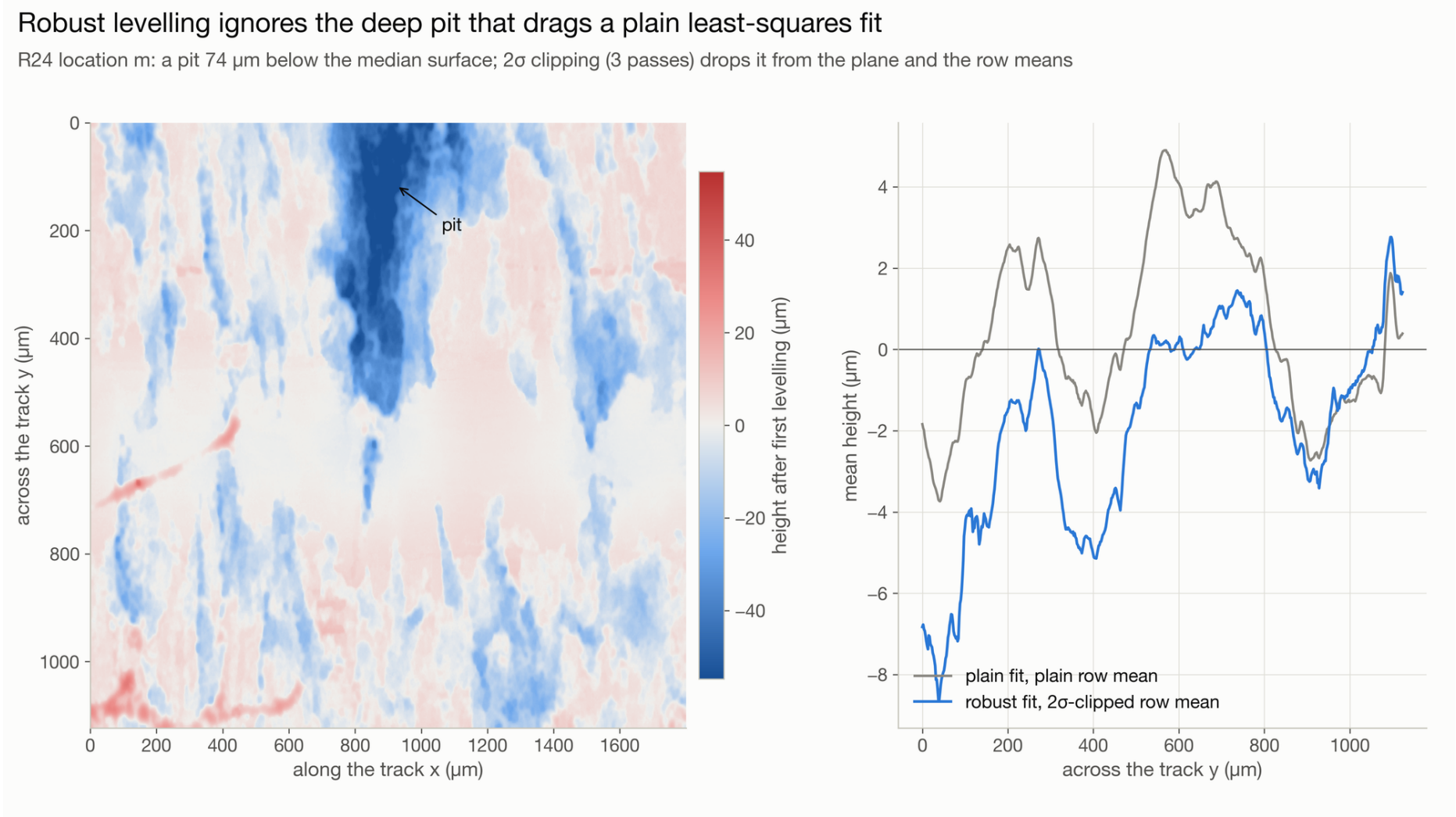
Data: outputs/data/per_cycle_cof.csv

Figure 5: Height map R04 (m) with core, band and reference rows; across-track profile with groove and pile-up shading.



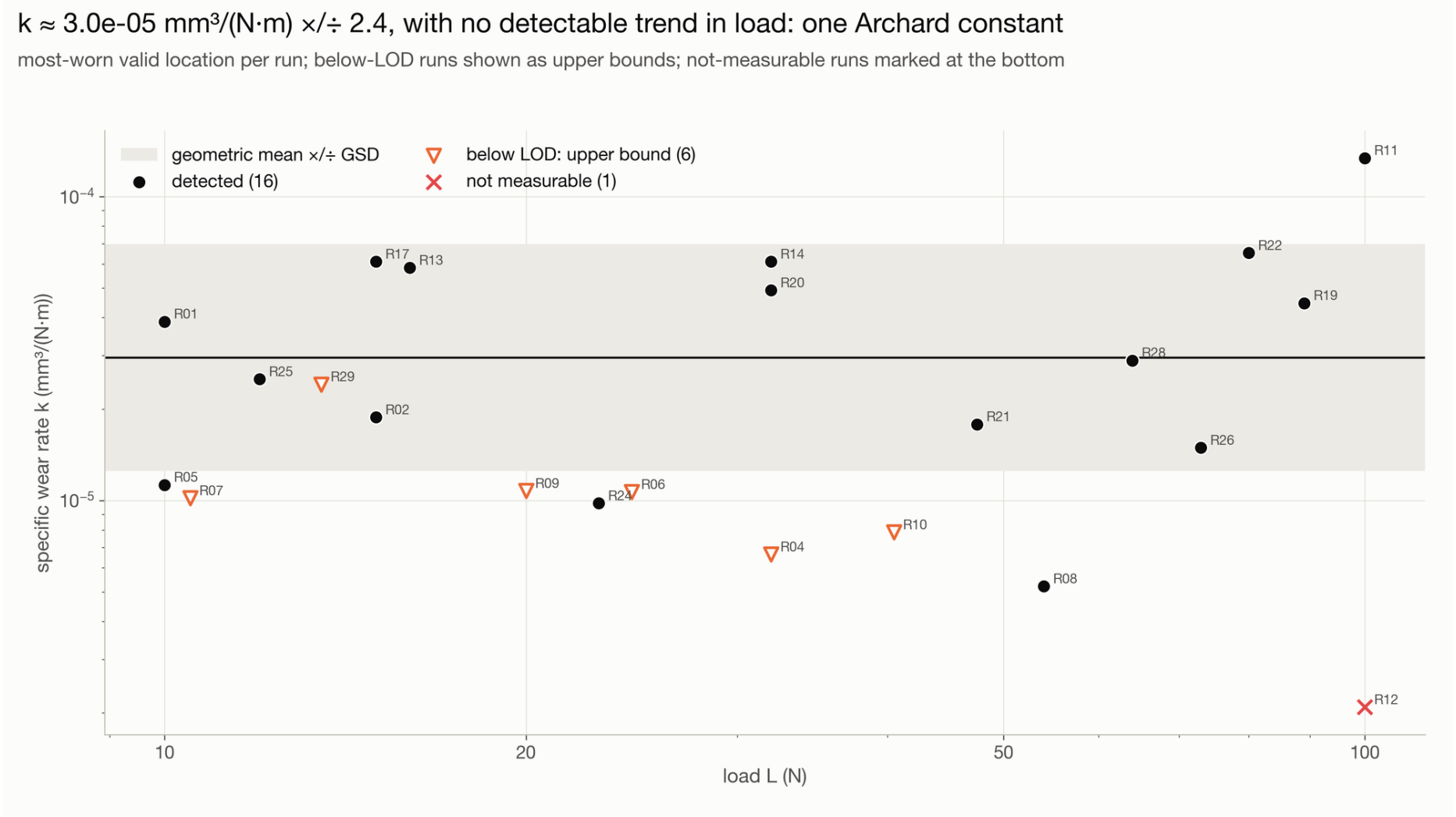
Data: testdata/R04/R04.bcrf

Figure 6: Robust vs plain levelling on R24 (m), a scan with a deep pit.



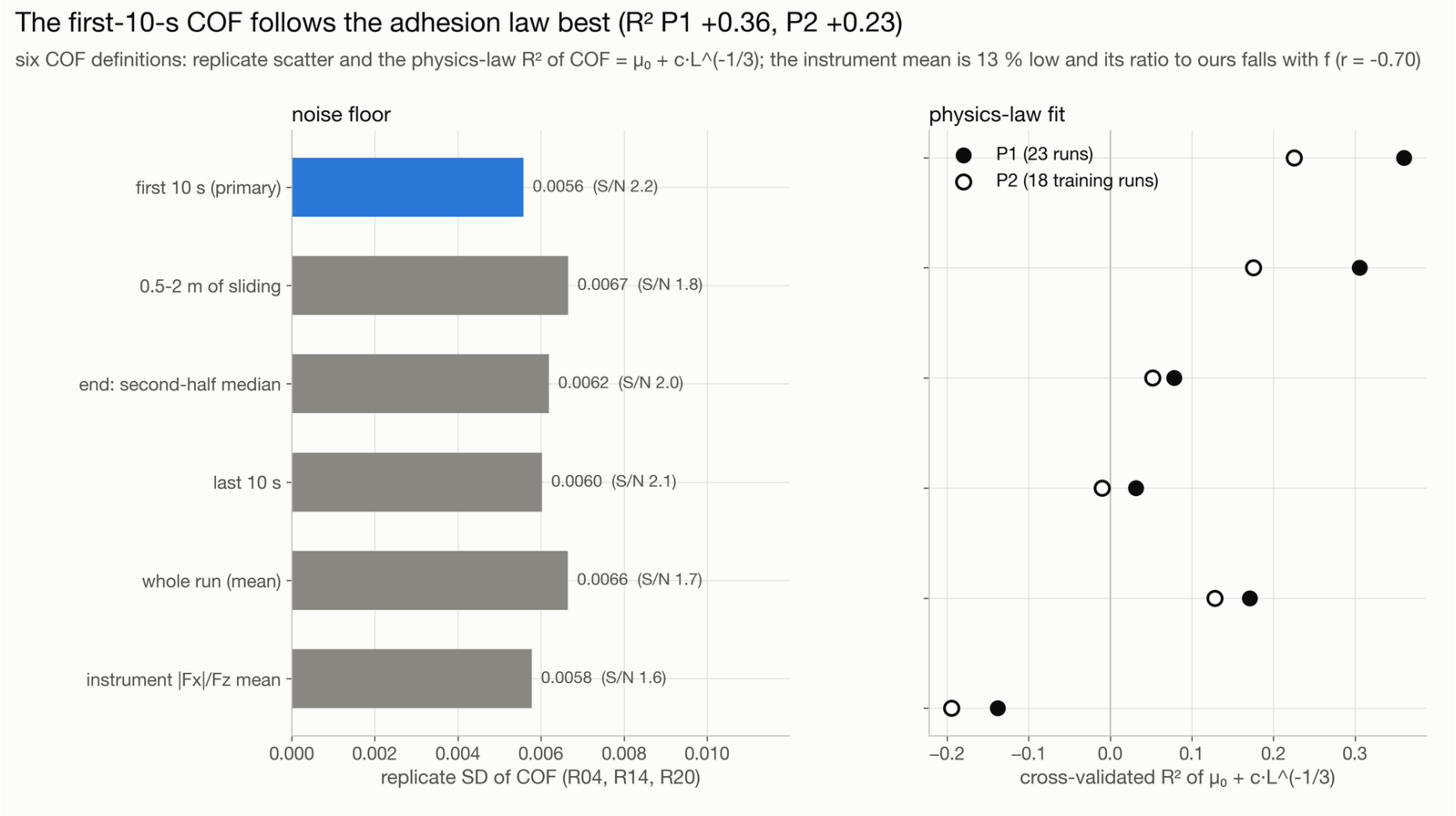
Data: testdata/R24/R24.bcrf

Figure 7: Specific wear rate k per run vs load with below-LOD upper bounds and not-measurable runs.



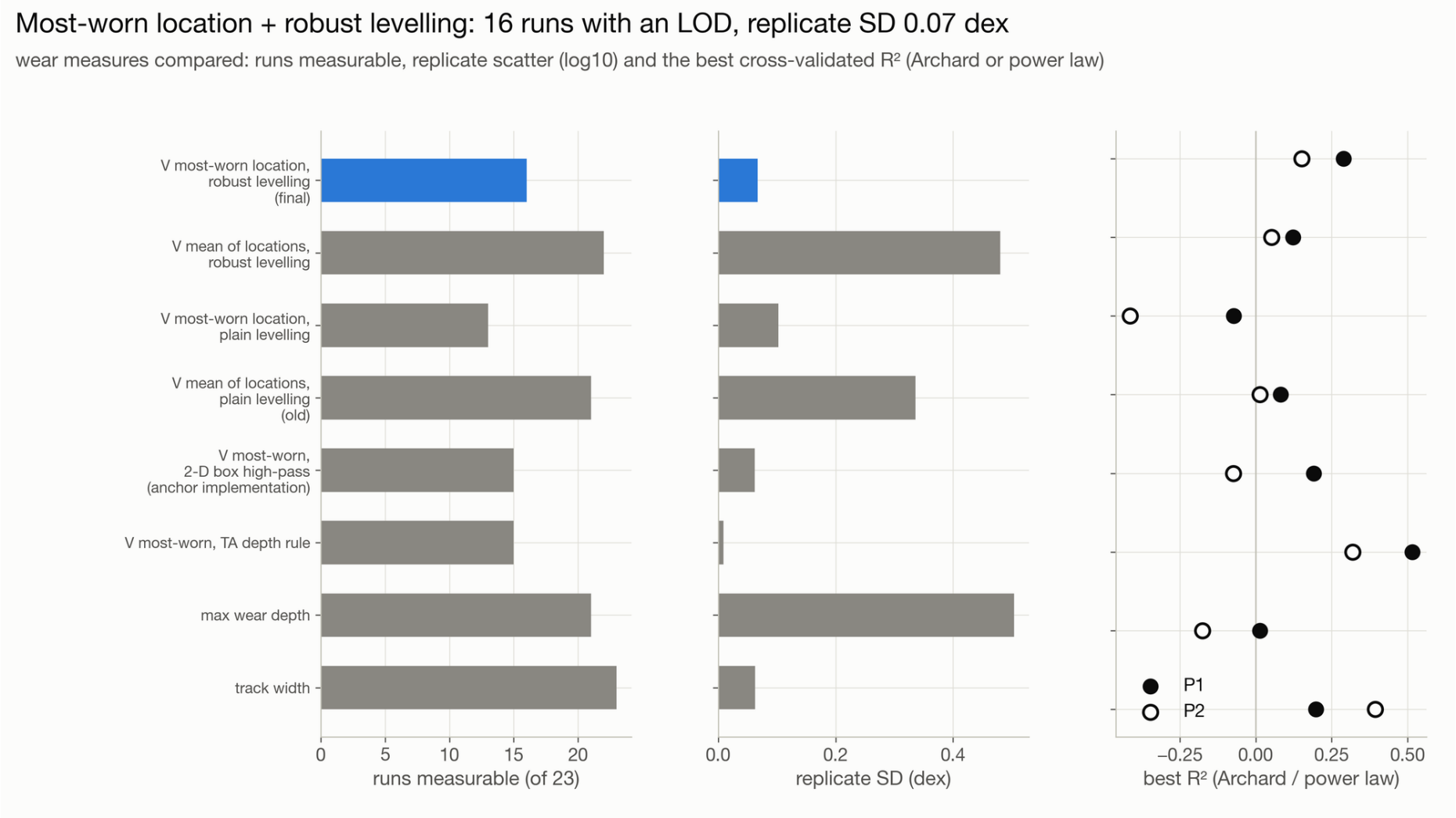
Data: outputs/data/runs.csv

Figure 8: C1: COF definitions compared by replicate SD and physics-law R^2 (P1, P2).



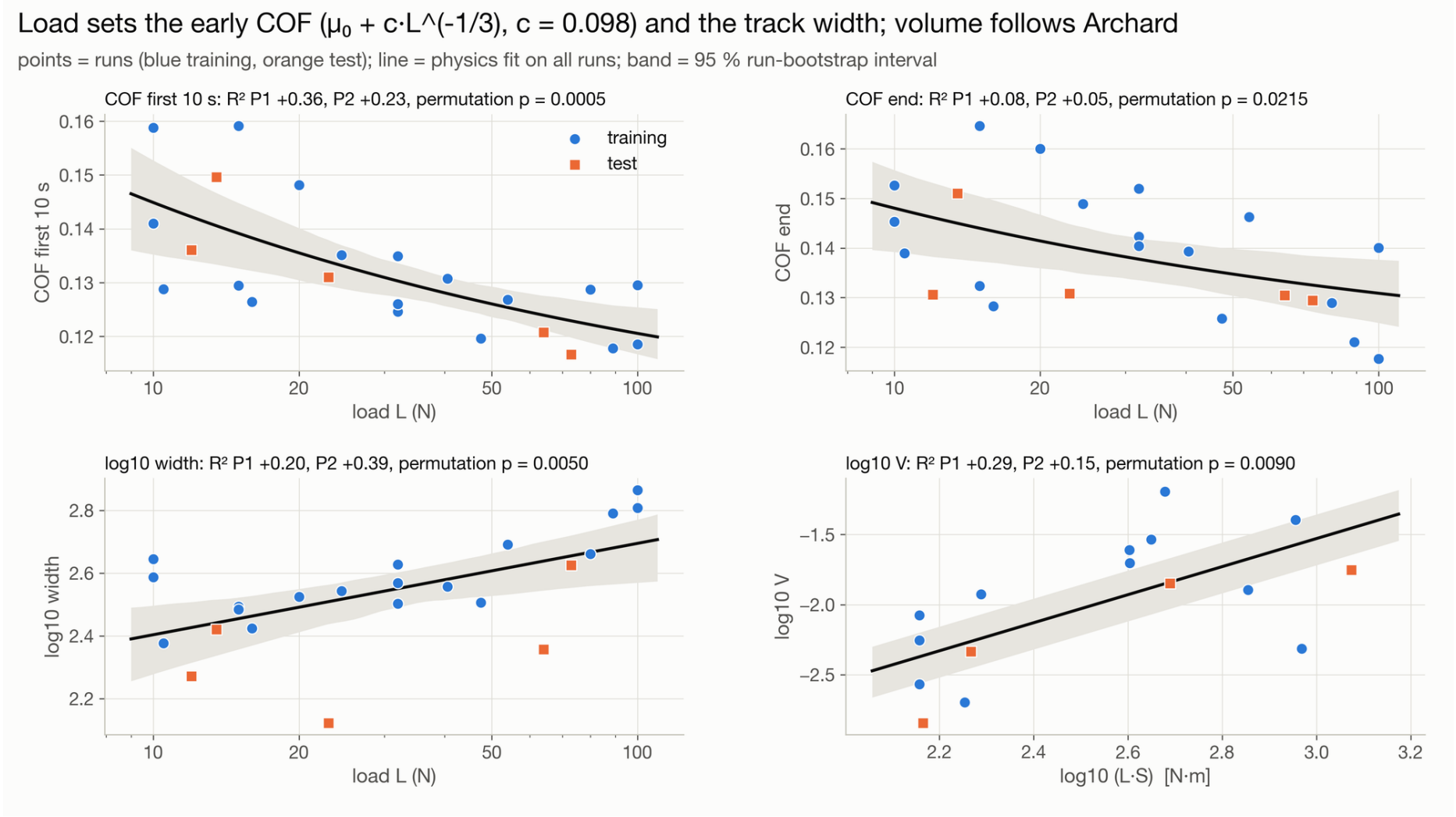
Data: outputs/tables/c1_cof_definitions.csv

Figure 9: C2: wear measures compared (runs measurable, replicate SD, best R^2 in P1/P2).



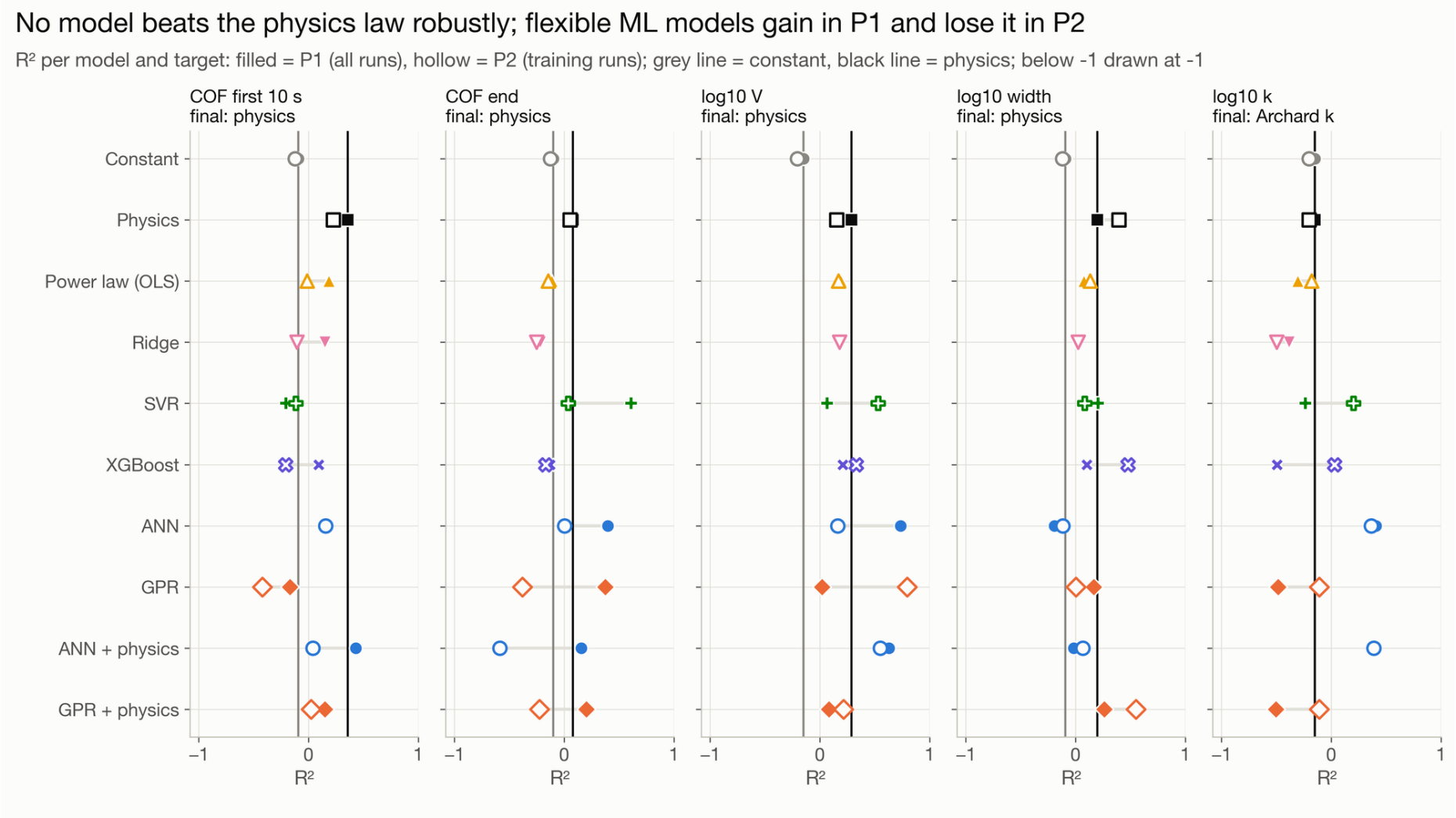
Data: outputs/tables/c2_wear_measures.csv

Figure 10: Physics fits: COF_first10s and COF_end vs load (μ_0 + c·L^(-1/3)), track width vs load (log-log), wear volume vs L·S (Archard, slope 1), each with a 95 % run-bootstrap band.



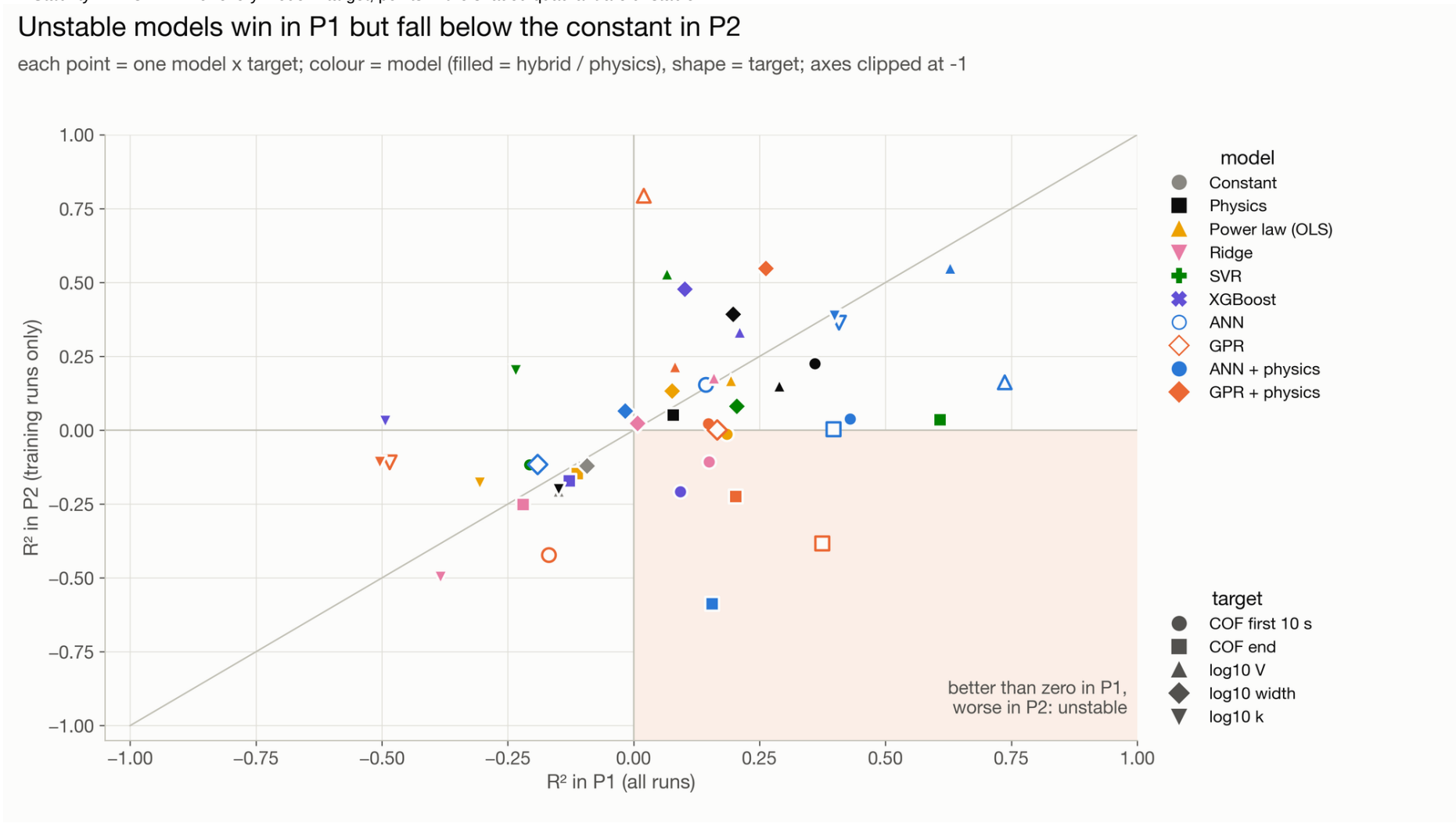
Data: outputs/data/runs.csv, outputs/tables/final_parameters.csv

Figure 11: C3: P1 (filled) and P2 (hollow) R^2 for every model and target, with constant and physics reference lines.



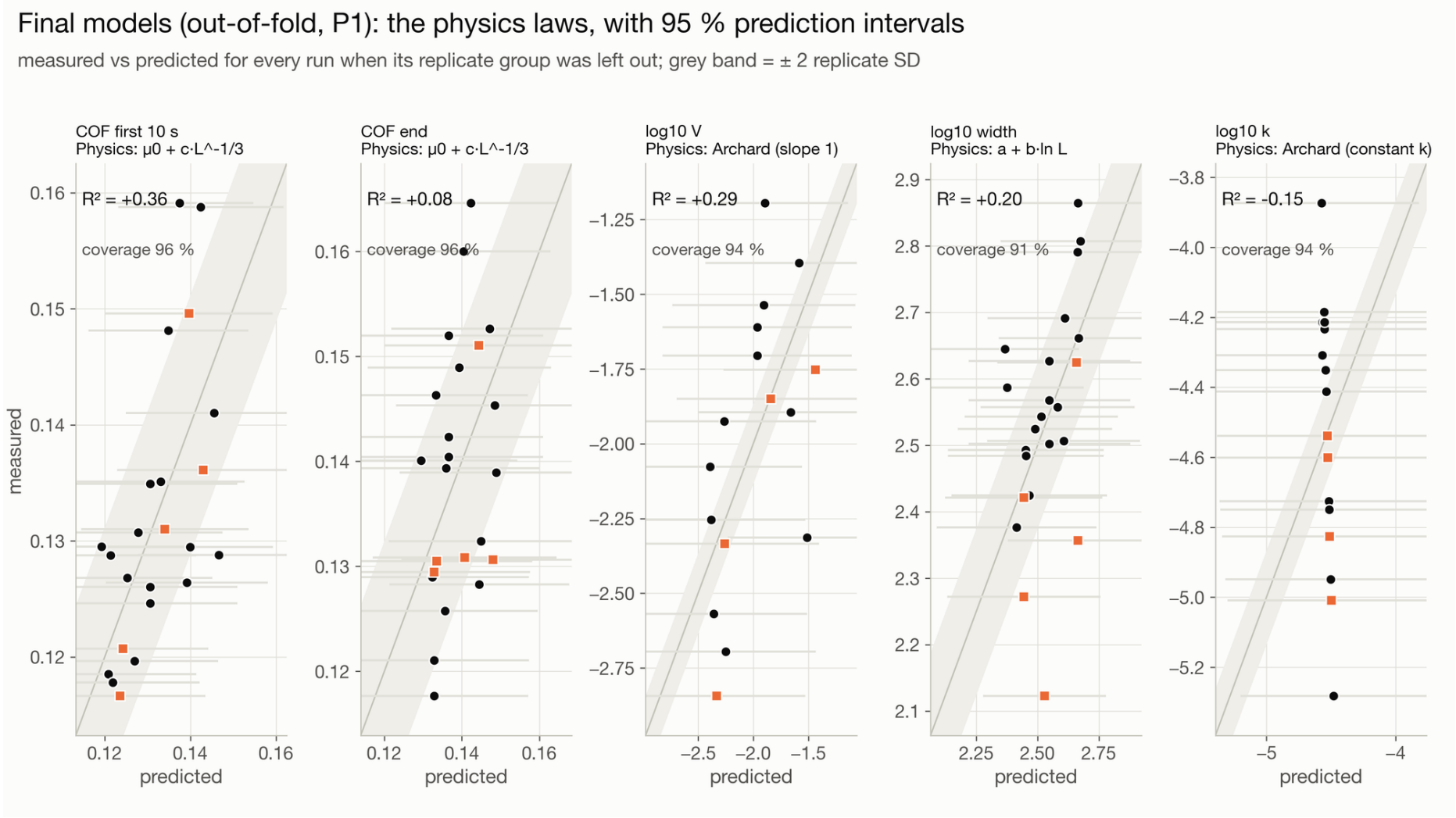
Data: outputs/tables/metrics_all.csv

Figure 12: Stability: P1 vs P2 R^2 for every model x target; points in the shaded quadrant are unstable.



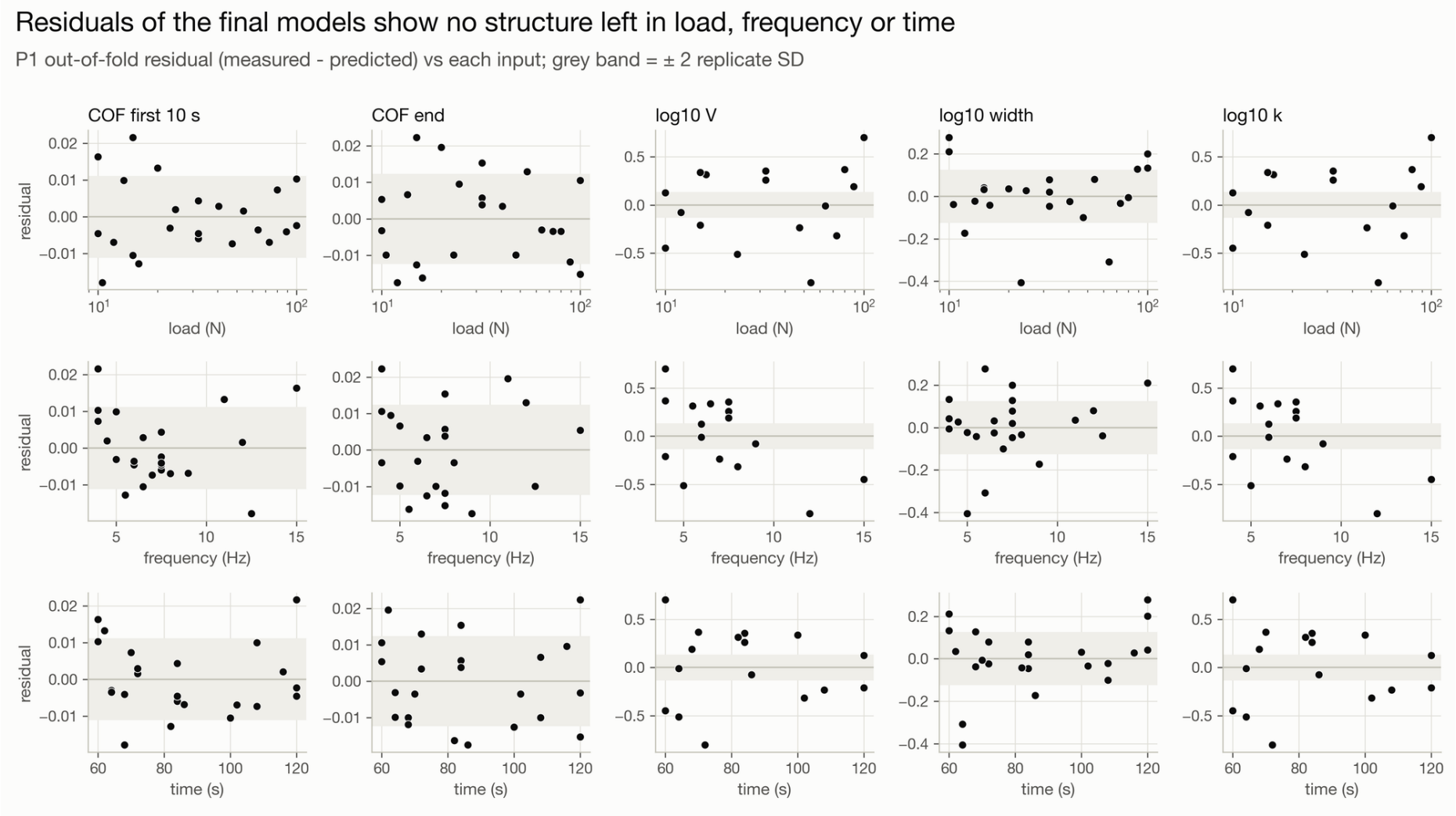
Data: outputs/tables/metrics_all.csv

Figure 13: Parity plots (P1 out-of-fold) of the final model per target with 95 % prediction intervals.



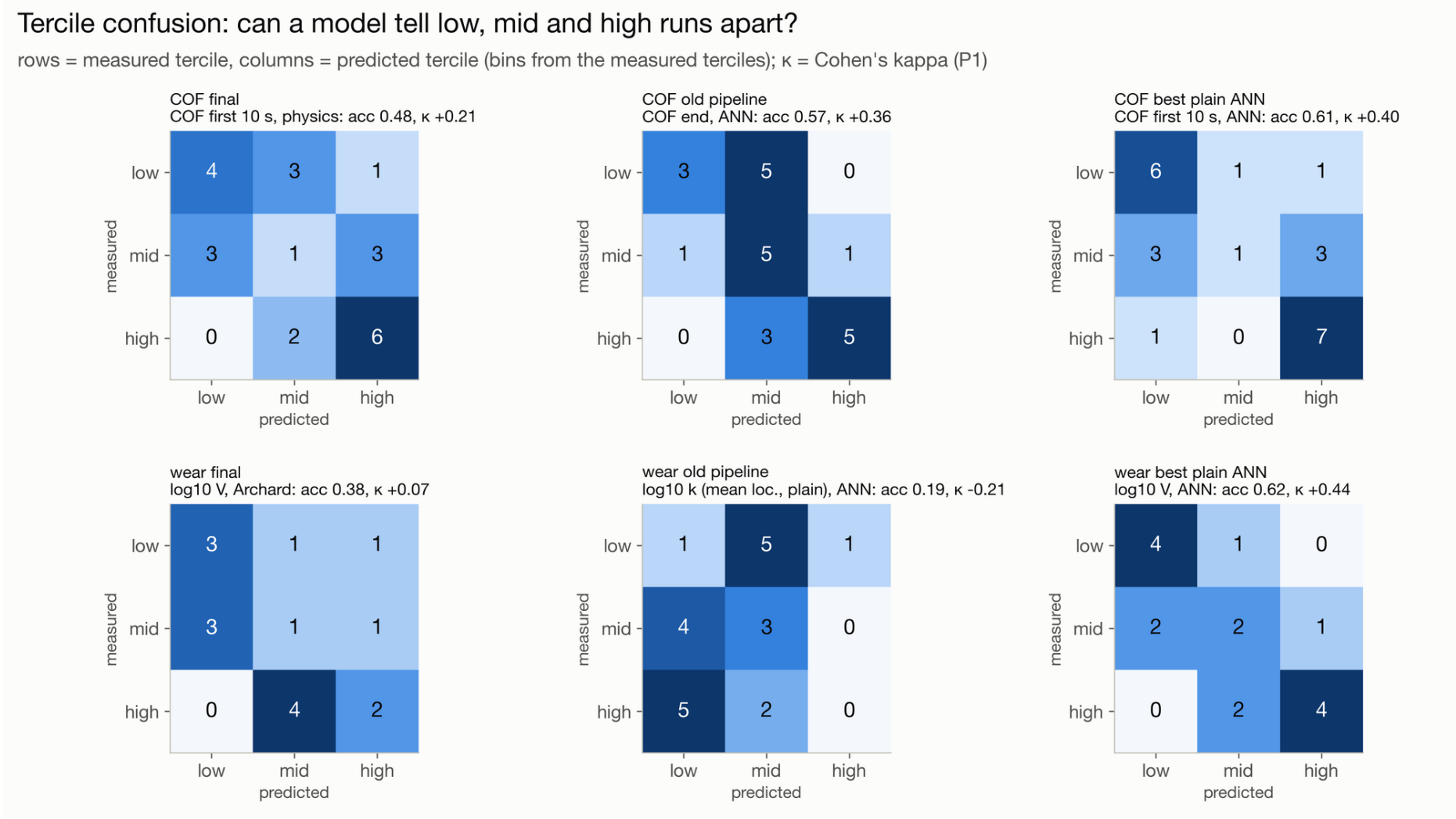
Data: outputs/data/oof_predictions.csv

Figure 14: Residuals of the final models vs load, frequency and time (P1 out-of-fold).



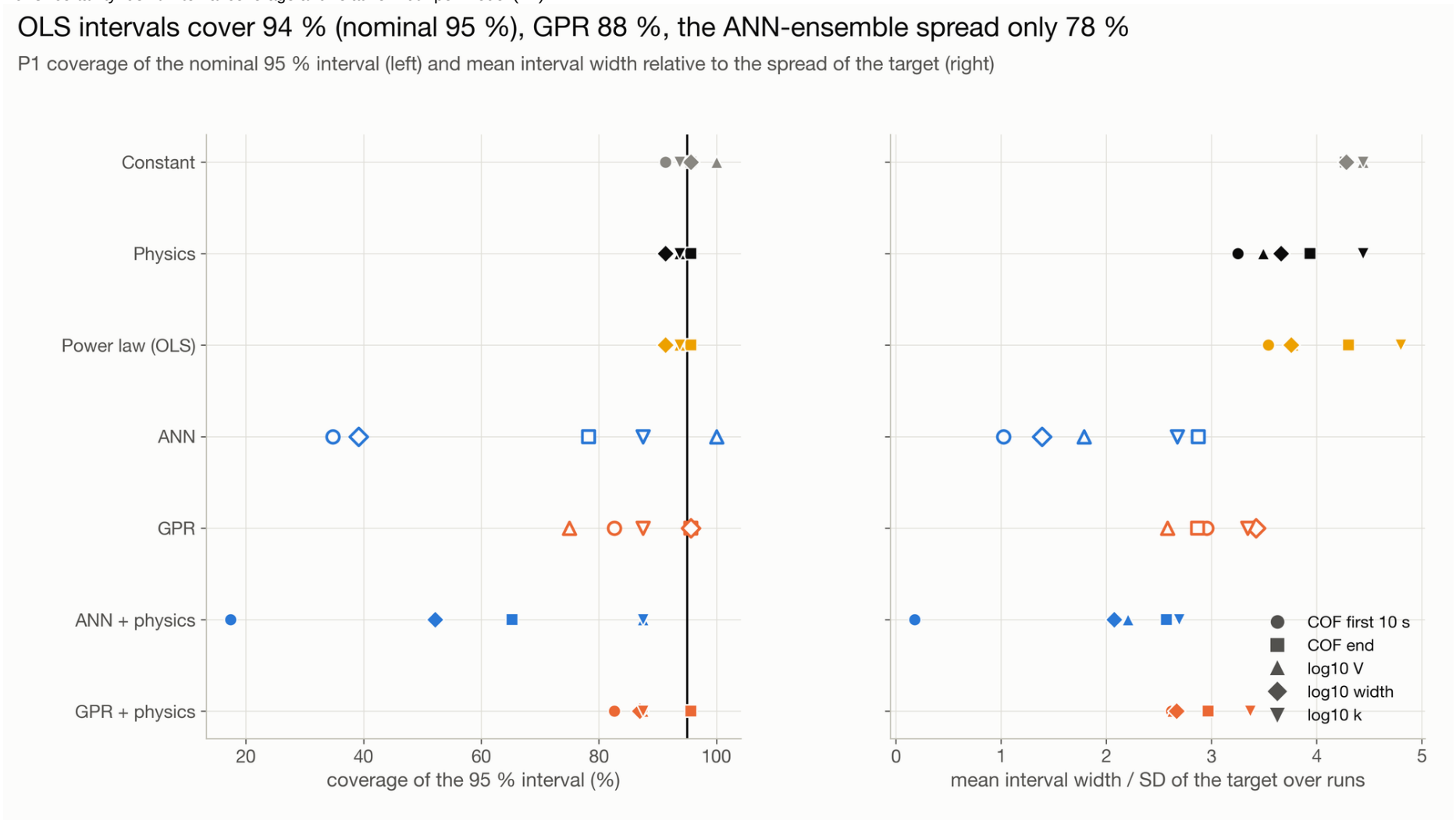
Data: outputs/data/oof_predictions.csv

Figure 15: Tercile confusion matrices (P1): final vs old pipeline vs best plain ANN, for COF and wear.



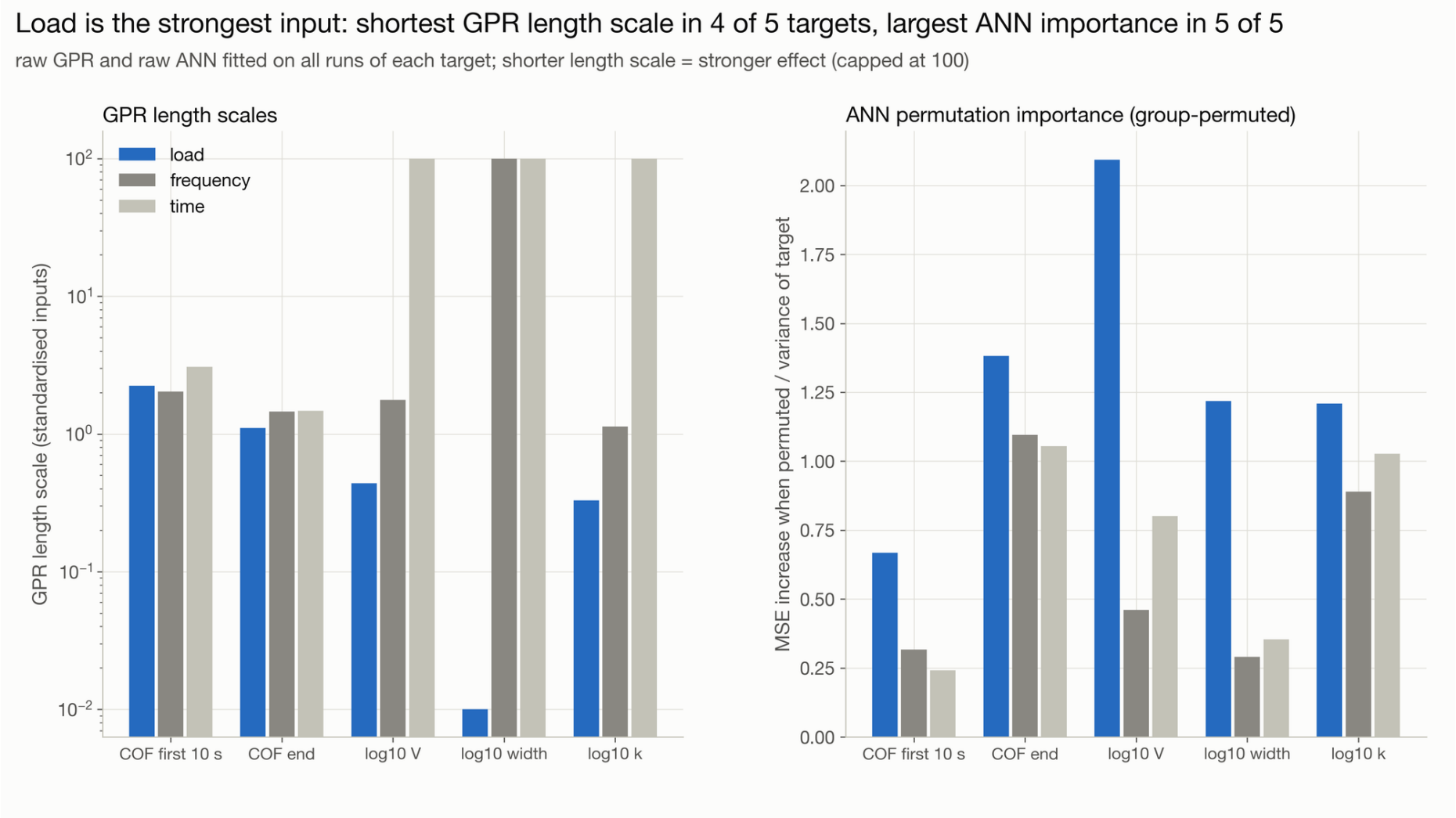
Data: outputs/data/oof_predictions.csv, outputs/data/c4_oof.csv

Figure 16: Uncertainty: 95 % interval coverage and relative width per model (P1).



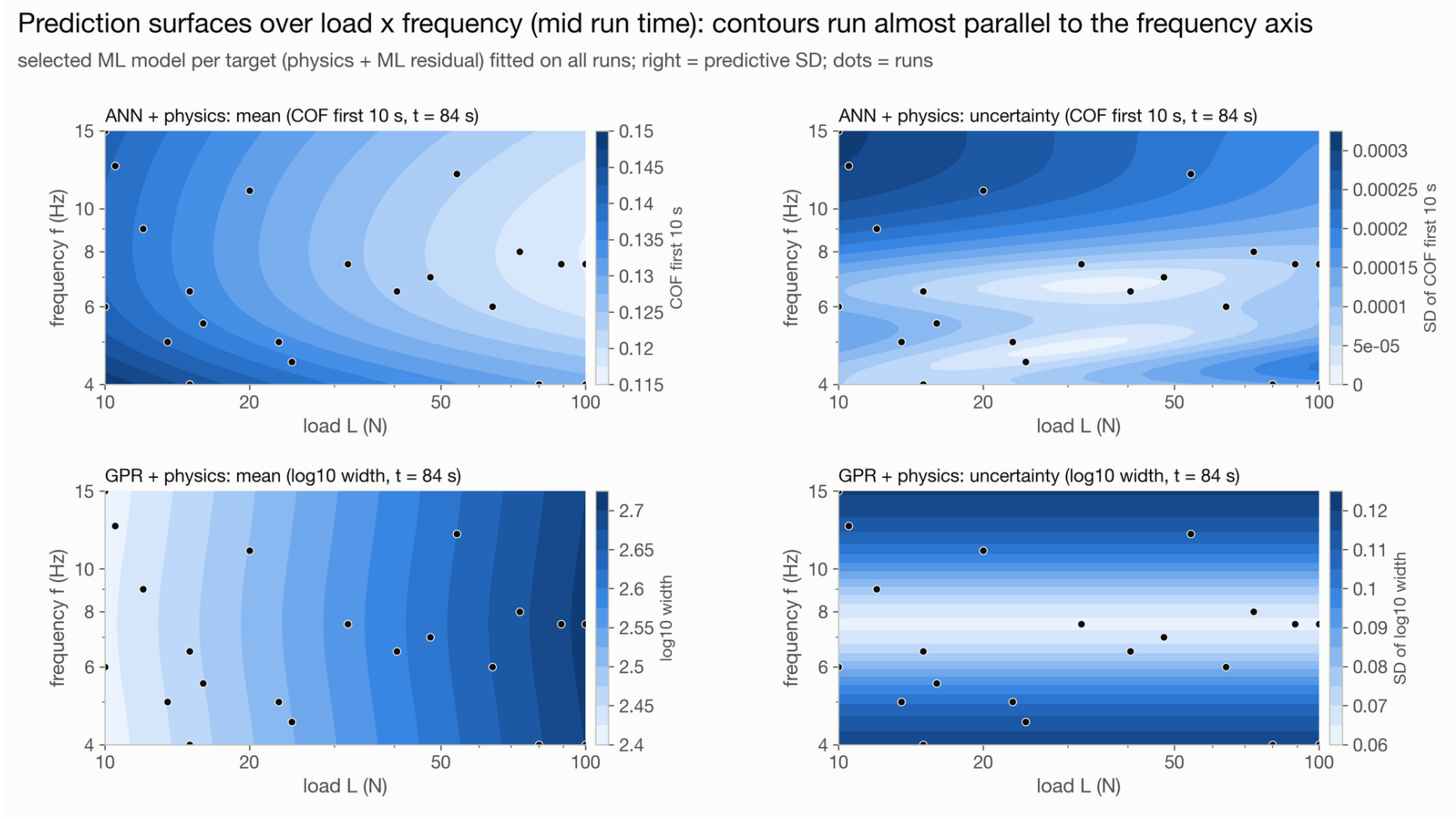
Data: outputs/tables/metrics_all.csv

Figure 17: Interpretation: GPR length scales and ANN permutation importance per input and target.



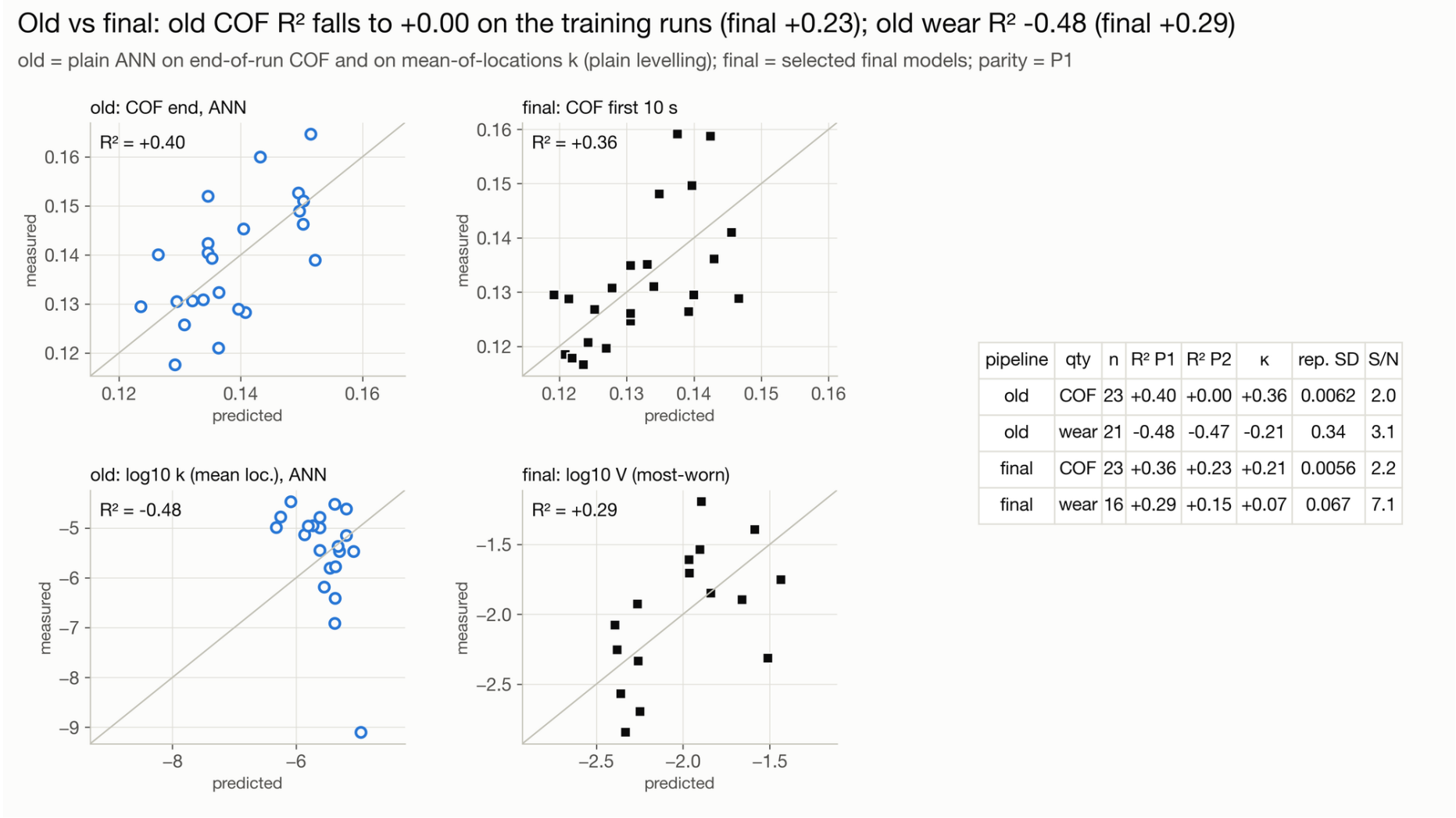
Data: outputs/tables/gpr_length_scales.csv, ann_permutation_importance.csv

Figure 18: Prediction surfaces (mean and SD) of the selected COF and width models over L x f at the median run time.



Data: outputs/data/runs.csv (models refitted on all runs)

Figure 19: C4: old pipeline vs final, metric table and P1 parity plots side by side.



pipeline	qty	n	R² P1	R² P2	κ	rep. SD	S/N
old	COF	23	+0.40	+0.00	+0.36	0.0062	2.0
old	wear	21	-0.48	-0.47	-0.21	0.34	3.1
final	COF	23	+0.36	+0.23	+0.21	0.0056	2.2
final	wear	16	+0.29	+0.15	+0.07	0.067	7.1

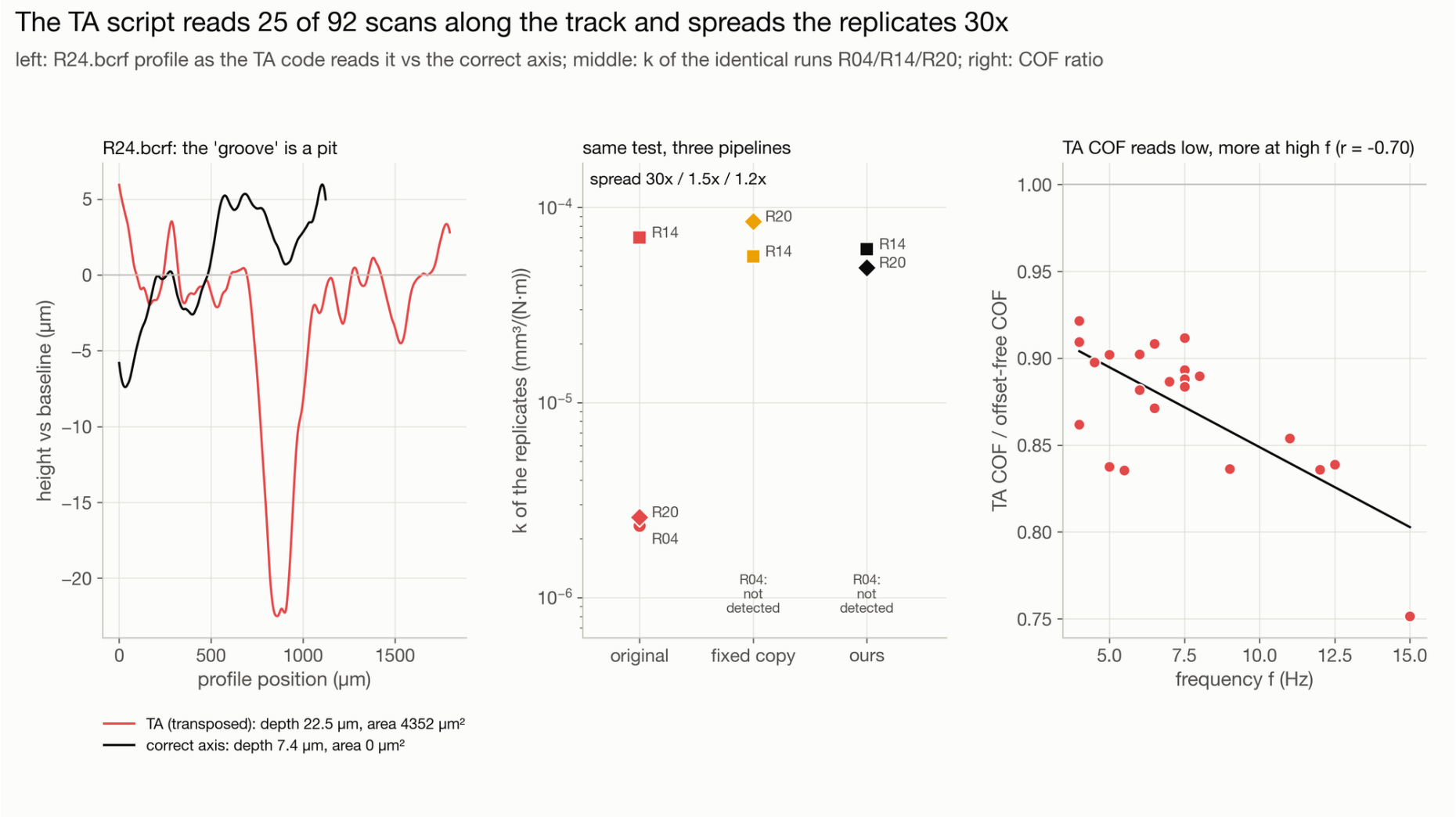
Data: outputs/tables/c4_old_vs_final.csv, outputs/data/c4_oof.csv

Figure 20: C5: relative COF and k vs load and speed, ours vs Ponnuswamy 2024, and the P1 scores of the pooling strategies.



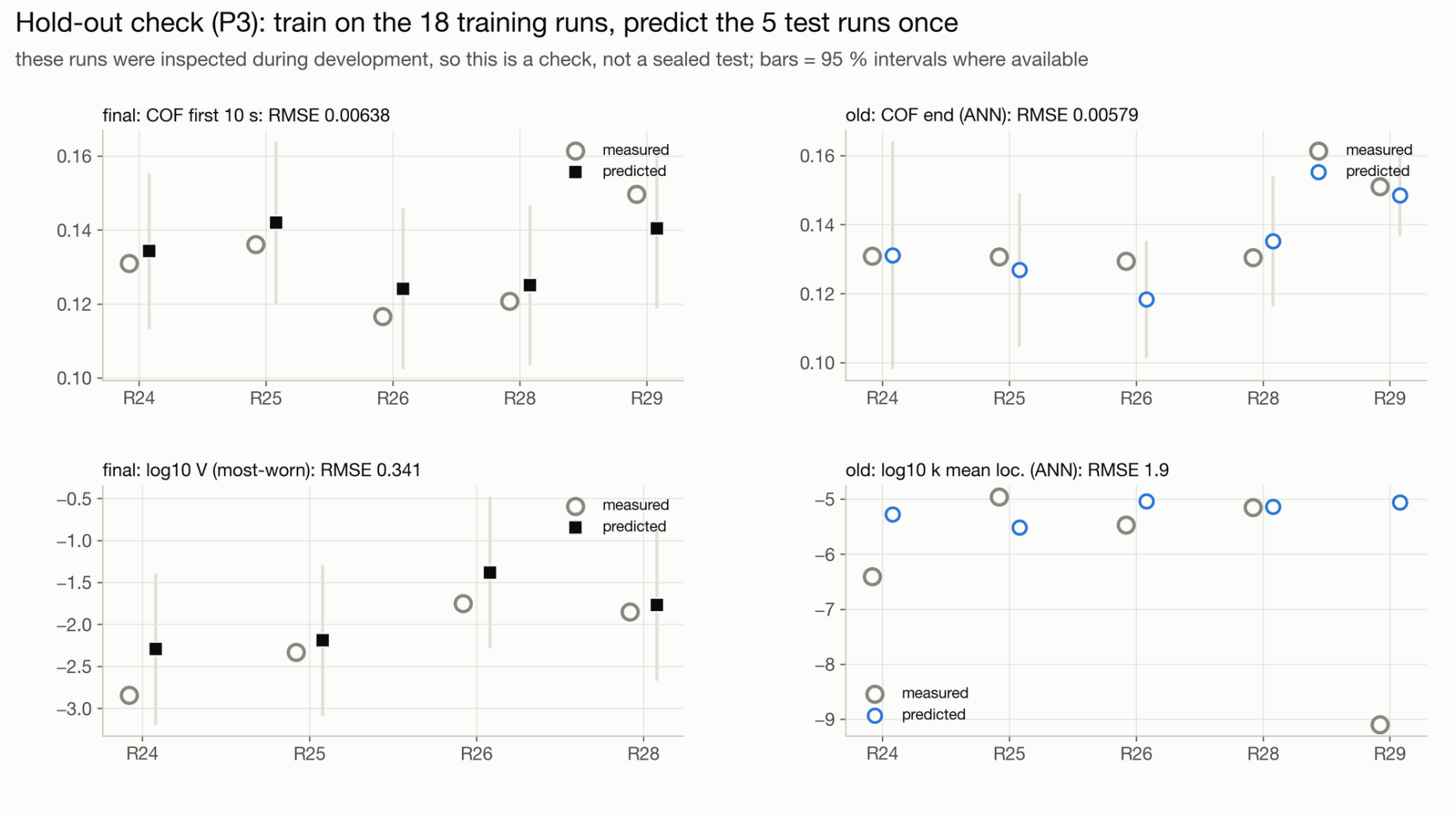
Data: inputs/ponnuswamy2024_table5.csv, outputs/tables/c5_trends.csv, c5_strategies.csv

Figure 21: C6: TA script - R24 orientation flip, replicate spread original/fixed/ours, COF ratio vs frequency.



Data: inputs/analyze_wear.py on testdata/, outputs/tables/c6_ta_*.csv

Figure 22: P3 hold-out: measured vs predicted for the 5 test runs, final vs old pipeline.



Data: outputs/data/c4_oof.csv